

PRACTICAL WEEK-2

Session-2:

Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved

First Program OUTPUT and OBSERVATION:

```
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week1$  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week1$ cd /mnt/d/OSSP_2520080069/Practical/Week2  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ ls  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ nano task1.c  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ echo "This is the content of File1." >  
File1.txt  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ cat File1.txt  
This is the content of File1.  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ gcc task1.c -o task1  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ ./task1  
File copied successfully!  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ cat File2.txt  
This is the content of File1.  
ananya@Ananya:/mnt/d/OSSP_2520080069/Practical/Week2$ █
```

The program successfully copied the contents of File1.txt to File2.txt using the open(), read(), write(), and close() system calls. The system calls provide an interface between the user program and the kernel, allowing the program to access and modify files managed by the operating system. The successful execution demonstrated basic file handling and the transition between user space and kernel space.

Second Program OUTPUT and OBSERVATION :

```
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libcpr2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=699056, ...}) = 0
mmap(NULL, 781112, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76d50f6c2000
mmap(0x76d50f6c2000, 507904, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3000) = 0x76d50f6c5000
mmap(0x76d50f741000, 176128, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7f000) = 0x76d50f741000
mmap(0x76d50f76c000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xa9000) = 0x76d50f76c000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76d50f76c000
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76d50f6db000
arch_prctl(ARCH_SET_FS, 0x76d50f6db000) = 0
set_tid_address(0x76d50f6be128) = 563
set_robust_list(0x76d50f6bde0, 24) = 0
rseq(0x76d50f6bcb0, 8x20, 0, 0x53053053) = 0
mprotect(0x76d50f6be000, 16384, PROT_READ) = 0
mprotect(0x76d50f76c000, 4096, PROT_READ) = 0
mprotect(0x76d50f892000, 4096, PROT_READ) = 0
mprotect(0x76d50f8c0000, 4096, PROT_READ) = 0
mprotect(0x76d50f8f1000, 4096, PROT_READ) = 0
mprotect(0x69d8c2943000, 1462272, PROT_READ) = 0
mprotect(0x76d50f93f000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
getrandom("/dev/urandom", 64, GRND_NONBLOCK) = 8
munmap(0x76d50f6f5000, 20887) = 0
statsf("/sys/fs/selinux", {f_type=SYSFS_MAGIC, f_bsize=4096, f_blocks=0, f_bfree=0, f_bavail=0, f_files=0, f_ffree=0, f_fsid={val=[0x64532b78, 0xbca775bf]}, f_namelen=255, f_frsize=4096, f_flags=ST_VALID|ST_NOSUID|ST_NODEV|TCPNFS|NODV|LID, ...}) = 0
statsf("/selinux", 0x7ff3c25f1a0) = -1 ENOENT (No such file or directory)
brk(NULL) = 0x6d08e7c2000
brk(0x6d08e7c41000) = 0x6d08e7c41000
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "nodev\tsysfs\nnodev\ttcpnfs\nnodev\tid....", 1024) = 442
lseek(3, 0, SEEK_CUR) = 442
lseek(3, 416, SEEK_SET) = 416
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0444, st_size=0, ...}) = 0
read(3, "none /usr/lib/modules/6.18.33.2....", 1024) = 1024
read(3, "e 0 0\nodevpts /dev/pts devpts rw,....", 1024) = 1024
read(3, ",access=client,msize=65536,trans....", 1024) = 1024
read(3, "des=1024,mode=708,noswap 0 0\ntmp....", 1024) = 262
read(3, "", 1024) = 0
close(3) = 0
access("/etc/selinux/config", F_OK) = -1 ENOENT (No such file or directory)
fcntl(0, F_GETFD) = 0
fcntl(1, F_GETFD) = 0
fcntl(2, F_GETFD) = 0
rt_sigaction(SIGPIPE, NULL, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
poll([{fd=0, events=0}, {fd=1, events=0}, {fd=2, events=0}], 3, 0) = 0 (Timeout)
rt_sigaction(SIGPIPE, {sa_handler=SIG_DFL, sa_mask=[PIPE], sa_flags=SA_RESTORER|SA_RESTART, sa_restorer=0x76d50f445cb0}, {sa_handler=SIG_DFL, sa_mask=[], sa_flags=0}, 0) = 0
```

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ananya@Ananya: /mnt/d/OSSP_2520080069/Practical/Week2$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffec99a5f48 /* 28 vars */) = 0
brk(NULL) = 0x6d08e7c20000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x76d50f8fa000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x76d50f8f5000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76d50f8c2000
mmap(0x76d50f8c9000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x76d50f8c9000
mmap(0x76d50f8ea000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x76d50f8ea000
mmap(0x76d50f8f1000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x76d50f8f1000
mmap(0x76d50f8f3000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x76d50f8f3000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76d50f894000
mmap(0x76d50f898000, 147456, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x76d50f898000
mmap(0x76d50f8bc000, 16384, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x76d50f8bc000
mmap(0x76d50f8c0000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2c000) = 0x76d50f8c0000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=1198376, ...}) = 0
mmap(NULL, 1200152, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76d50f76e000
mmap(0x76d50f780000, 655360, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x12000) = 0x76d50f780000
mmap(0x76d50f820000, 466944, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xb2000) = 0x76d50f820000
mmap(0x76d50f892000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x123000) = 0x76d50f892000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0....", 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 840, 64) = 840
fstat(3, {st_mode=S_IFREG|0755, st_size=2186512, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0....", 840, 64) = 840
mmap(NULL, 2231696, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x76d50f400000
mmap(0x76d50f420000, 1671168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20000) = 0x76d50f420000
mmap(0x76d50f5c0000, 319488, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x76d50f5c0000
mmap(0x76d50f60e000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x20d000) = 0x76d50f60e000
mmap(0x76d50f614000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x76d50f614000
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